Junseong Ahn

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Portfolio · LinkedIn · Google Scholar
Seoul, South Korea

Applied AI engineer focused on embodied perception, autonomous systems, and real-world deployment of multimodal intelligent systems.

Research Interests

Embodied AI, Autonomous Systems, Multimodal Perception, 3D Computer Vision, Sensor Fusion, Generative Models, Real-Time AI Systems, Edge AI Deployment

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Education

Korea University — M.S. in Computer Science, GPA: 3.82/4.0
Seoul, South Korea | Jun 2024
Brigham Young University — B.S. in Applied & Computational Mathematics, GPA: 3.59/4.0
Provo, UT | Apr 2020

Professional Experience

Deep Learning Engineer — Imagoworks

Seoul, South Korea | Aug 2020 – Oct 2023

- Developed and optimized 3D deep learning systems for browser-based and edge AI deployment using WebAssembly and ONNX
- Designed and deployed automated 3D scan processing pipelines for clinical AI workflows
- Reduced inference memory usage for browser-based 3D models enabling real-time edge deployment
- Designed AI-driven data pipelines for scalable medical imaging workflows
- Contributed to 2 journal publications and 10+ domestic and international patent applications

Technologies: C/C++ (VTK, WebAssembly), Python (PyTorch, OpenCV, ONNX), JavaScript (React)

Machine Learning Engineer Intern — Navier Boat

San Mateo, CA | Mar 2020 – Aug 2020

- Developed embodied perception systems for autonomous surface vehicles in real-world maritime environments
- Built multi-sensor perception pipelines integrating radar, vision, and Kalman filter-based tracking
- Processed and managed ~2TB/day of sensor and video data using AWS S3 infrastructure
- Developed object detection models under limited-data conditions for real-world deployment
- Demonstrated autonomous navigation systems to industry experts in San Francisco Bay

Technologies: Python (PyTorch, OpenCV), C/C++, CUDA, ROS, AWS

Research Experience

Deep Learning Researcher — Perception, Control, and Cognition Lab (PCCL), Brigham Young University

Provo, UT | Feb 2019 – Apr 2020

- Collaborated with researchers and medical professionals from India and the University of Michigan to develop AI-assisted cancer screening systems
- Developed graphical representations for analogical reasoning in word embeddings

- Designed and evaluated 20+ algorithms for analogy reasoning and representation learning
- Improved deep learning experimentation efficiency by containerizing research environments using Docker
- Awarded 1st Place at the Student Research Conference

Technologies: Python (PyTorch), Docker, Parallel Processing, Computer Vision

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Algorithm Researcher — Impact Lab, Brigham Young University

Provo, UT | May 2019 – Apr 2020

- Developed optimization algorithms for large-scale transportation routing systems
- Conducted large-scale experiments comparing clustering algorithms against baseline K-means methods
- Led preprocessing and analysis of political district datasets exceeding 100GB

Technologies: Python (Scikit-learn, Pandas, GeoPandas)

Publications

Choi, J., **Ahn, J.**, Park, J.
"Deep Learning-based Automated Detection of the Dental Crown Finish Line"
Journal of Prosthetic Dentistry, 2024

Cho, J., Yi, Y., Choi, J., **Ahn, J.**, Yoon, H., Yilmaz, B.
"Time Efficiency, Occlusal Morphology, and Internal Fit of Anatomic Contour Crowns Designed by Dental Software Powered by Generative Adversarial Networks: A Comparative Study"
Journal of Dentistry, 2023

Selected Patent Applications

- 3D Facial Scan–Medical Image Registration — PCT/KR2024/095421
- Automated 3D Scan Alignment — PCT/KR2024/011839
- 3D Scan-Based Prosthesis Generation — PCT/KR2024/005369
- Automated Prosthesis Recommendation from 3D Oral Data — PCT/KR2024/008491
- AI-based Prosthesis Generation using Panoramic Images — PCT/KR2023/006810

Conference Presentations

- "AI-based Dental Prosthesis Design Process" — International Dental Show, 2024
- "Efficient Dental Prosthesis Design with AI" — Society for Computational Design and Engineering, 2024

Teaching Experience

Math Tutor — Brigham Young University
Provo, UT | Jan 2019 – Mar 2019

- Taught advanced mathematics and problem-solving strategies to undergraduate students
- Prepared structured lesson plans tailored to individual academic goals

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Honors & Awards

- 1st Place, Student Research Conference — Attention-Based Cervical Cancer Detection (2020)
- Don Robinson Scholarship for Academic Excellence in Mathematics (2019)
- BYU Academic Scholarship (2019)

Leadership & Service

Full-Time Missionary — The Church of Jesus Christ of Latter-day Saints

Busan Korean Mission | Dec 2024 – Apr 2026

- Provided leadership, mentoring, and community outreach through full-time volunteer service

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Technical Skills

Programming: Python, C/C++, JavaScript
Machine Learning & Vision:
PyTorch, OpenCV, ONNX, Deep Learning, Computer Vision
Systems & Robotics:
ROS, CUDA, Docker, WebAssembly
Cloud & Data:
AWS (S3), Large-Scale Data Pipelines

Languages

- Korean — Native
- English — Fluent

References

Available upon request